INT428: AI - ESSENTIALS
PROJECT REPORT
(Project Semester January-May 2026)

AI Fashion Stylist (AI-FS)

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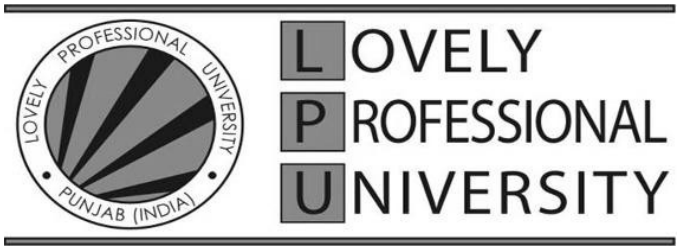
Programme and Section: B.Tech 424SA
Course Code: INT428
Roll No: 54

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DECLARATION

I, Akash Kumar, student of B.Tech under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 4 May 2026

Registration No. 12416601

_____

Signature
Akash Kumar
Name of the Student

CERTIFICATE

This is to certify that Akash Kumar bearing Registration No. 12416601 has completed INT428 project titled "AI Fashion Stylist (AI-FS) — An Intelligent Outfit Recommendation System using Generative AI" under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Mr. Atul Kumar

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ACKNOWLEDGEMENT

I express my sincere gratitude to my faculty supervisor Atul Kumar for his continuous guidance, valuable suggestions, and constant encouragement throughout the completion of this project. His expertise in Artificial Intelligence and Web Development was instrumental in shaping the direction and quality of this work.

I am also grateful to the Department of CSE, Lovely School of AI & Emerging Technologies, Lovely Professional University, for providing the necessary resources and infrastructure to complete this project.

Special thanks to the open-source community behind Next.js, Groq SDK, SQLite, and Lucide React for making state-of-the-art tools accessible to developers.

Akash Kumar

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AI Fashion Stylist (AI-FS)

Lovely Professional University | Akash Kumar | 12416601

1. PROJECT OVERVIEW

1.1 PROBLEM STATEMENT

Design and implement an intelligent fashion recommendation web application that enables users to receive personalized AI-generated outfit suggestions based on their occasion and preferred style. The system should incorporate a conversational AI powered by large language models and provide a seamless authenticated multi-user experience, making it possible for users to manage, save, and revisit their outfit recommendations.

1.2 PROJECT GOALS

AI Fashion Stylist (AI-FS) is an intelligent, full-stack fashion recommendation system built with Next.js 16 and TypeScript. The project addresses a fundamental challenge faced by everyday users: the difficulty of deciding what to wear for a specific occasion or matching outfits to a personal aesthetic. This system implements a complete AI-powered recommendation pipeline that accepts user context (occasion and style preference), queries a large language model via the Groq SDK (LLaMA 3.3-70B), and returns a structured, stylish outfit description.

Additionally, the system integrates a secure multi-user authentication mechanism using JWT and bcrypt, a persistent SQLite database for outfit history, and a conversational fashion assistant interface, alongside a clean and responsive UI built with Lucide React icons.

1.3 EXPECTED OUTCOMES

- A fully functional Next.js web application for outfit recommendation, history tracking, and AI chat
- Secure user authentication using JWT tokens and bcrypt password hashing
- AI outfit recommendation engine powered by Groq's LLaMA 3.3-70B-Versatile model
- Persistent outfit history stored in a SQLite database per authenticated user
- Conversational fashion assistant for follow-up queries and style advice
- Responsive UI with Lucide React icons and modern Next.js App Router architecture

1.4 SCOPE

Table 1 — Project Scope (In Scope vs Out of Scope)

Area	In Scope	Out of Scope
User Auth	JWT login/register; bcrypt hashing; token-based session management	OAuth, SSO, biometric login
AI Recommendation	Groq LLaMA 3.3-70B; occasion + style input; structured outfit output	Fine-tuning, image-based style detection

Database	SQLite; outfits table per user; history retrieval	Cloud DB, multi-tenant SaaS, Redis cache
Chat Assistant	Conversational Q&A with Groq API; session-based history	Voice input, real-time streaming, multi-user sessions
Frontend	Next.js 16 App Router; TypeScript; Lucide React icons	React Native, mobile app, PWA
Deployment	Local Next.js dev server; environment variables for secrets	Docker, Kubernetes, CI/CD pipeline

2. MODULE-WISE BREAKDOWN

AI-FS is architected into four clearly separated modules. This architecture ensures the authentication logic is independently testable, the Next.js App Router UI is a pure presentation layer, and the AI recommendation and chat modules respond to every user request through a clean API pipeline — no module reaches into another module's internals directly.

Table 2 — Module Summary

Module	Name	Technology	Primary Responsibility
M1	User Authentication & Profile	bcryptjs, jsonwebtoken, SQLite	Register, login, JWT issuance, session guard
M2	AI Outfit Recommendation	Groq SDK, LLaMA 3.3-70B	Occasion + style input → structured outfit generation
M3	Outfit History & DB Management	SQLite, sqlite3, Next.js API	Save, retrieve, display past outfit recommendations
M4	Conversational Fashion Assistant	Groq SDK, LLaMA 3.3-70B	Multi-turn fashion chat with session history

2.1 MODULE 1 — USER AUTHENTICATION & PROFILE MANAGEMENT

2.1.1 Purpose

- Accept user registration requests with username and plaintext password via a Next.js API route
- Hash passwords using bcryptjs before storage; compare hashed passwords on login
- Issue signed JWT tokens on successful authentication; store in HTTP-only cookies or client state
- Guard protected API routes by verifying JWT tokens on every request using jsonwebtoken
- Store user records in the SQLite 'users' table with auto-incremented IDs

2.1.2 Role in the Project

The User Authentication module is the security gateway for the entire AI-FS pipeline. Every personalized feature — outfit recommendations, history retrieval, and chat sessions — is gated behind a valid JWT. By using bcryptjs with a cost factor of 10 for password hashing and jsonwebtoken with a

secret stored in environment variables, the system ensures that user credentials are never stored in plaintext and that session tokens are tamper-proof.

2.2 MODULE 2 — AI OUTFIT RECOMMENDATION ENGINE

2.2.1 Purpose

- Accept user inputs: occasion (e.g., 'office', 'wedding', 'gym') and style preference (e.g., 'casual', 'formal', 'streetwear')
- Construct a structured prompt and call the Groq SDK with the LLaMA 3.3-70B-Versatile model
- Parse and return a complete outfit description including top, bottom, footwear, accessories, and colour palette
- Save the generated outfit to the SQLite 'outfits' table linked to the authenticated user's ID
- Display the recommendation on the frontend with clear styling and icon support via Lucide React

2.2.2 Role in the Project

The AI Outfit Recommendation Engine is the intelligence core of AI-FS. It transforms vague style intent into actionable, detailed outfit guidance. By leveraging the LLaMA 3.3-70B model through Groq's ultra-low-latency inference API, the system delivers responses in under two seconds. The structured prompt engineering approach ensures that outputs are consistently formatted, covering all components of an outfit rather than producing generic or incomplete advice.

2.3 MODULE 3 — OUTFIT HISTORY & DATABASE MANAGEMENT

2.3.1 Purpose

- Store every accepted outfit recommendation in the SQLite 'outfits' table with user_id, occasion, style, description, and created_at timestamp
- Provide an authenticated API route to retrieve the outfit history for the logged-in user, ordered by most recent
- Display the history in a scrollable list on the frontend with occasion, style tag, and full description
- Support future deletion and filtering operations through the structured relational schema

2.3.2 Role in the Project

The Outfit History module gives AI-FS a memory. Without persistent storage, every recommendation would be ephemeral, preventing users from revisiting looks they previously liked. By linking every outfit record to a user_id foreign key in SQLite, the system maintains strict data isolation between users — one user can never see another's history. The CURRENT_TIMESTAMP default on created_at enables chronological ordering without additional application logic.

2.4 MODULE 4 — CONVERSATIONAL FASHION ASSISTANT (CHAT)

2.4.1 Purpose

- Accept free-form fashion questions from the user (e.g., 'What shoes go with a navy suit?')

- Maintain a session-based conversation history to enable contextually coherent multi-turn dialogue
- Send the full conversation history plus the new user message to Groq's LLaMA 3.3-70B model
- Display the AI response with clear attribution and scrollable conversation UI

2.4.2 Role in the Project

The Conversational Fashion Assistant transforms AI-FS from a single-shot recommendation tool into an interactive style advisor. Users can ask follow-up questions, request alternatives, or explore style reasoning in natural conversation. Unlike the recommendation engine — which operates on structured inputs — the chat module handles open-ended queries, making it the most flexible and exploratory interface in the system. Session history is managed in client-side state per Next.js convention for stateless server components.

3. FUNCTIONALITIES

3.1 MODULE 1 — AUTHENTICATION FEATURES

Table 3 — Authentication Module Key Features

Feature	Description	Example
User Registration	POST /api/auth/register; hashes password with bcryptjs cost 10; inserts user row in SQLite	Username 'Akash' + password → stored as bcrypt hash
User Login	POST /api/auth/login; compares submitted password against stored hash; issues signed JWT	Correct password → JWT token returned to client
JWT Guard	Middleware verifies Bearer token on protected routes using jsonwebtoken.verify()	Missing or expired token → 401 Unauthorized response
Password Security	bcryptjs.hash() with saltRounds=10; bcryptjs.compare() for verification	Plaintext never stored; brute-force resistant
Session State	JWT payload contains userId and username; decoded on every API call	Frontend reads token to display logged-in user's name

3.2 MODULE 2 — OUTFIT RECOMMENDATION FEATURES

Table 4 — AI Recommendation Module Key Features

Feature	Description	AI Concept Demonstrated
Occasion Input	Dropdown or text field accepting occasion type (office, wedding, gym, date, casual)	Controlled natural language slot filling
Style Preference	User selects or types style (formal, casual, streetwear, bohemian, etc.)	User intent disambiguation for prompt construction
Groq API Call	groq.chat.completions.create() with structured system prompt and user context	LLM inference via API; few-shot prompt engineering
Outfit Parsing	Response includes top, bottom, footwear, accessories, colours as structured output	Prompt-constrained structured generation
DB Persistence	Accepted outfit saved to SQLite outfits	Relational data model; foreign key

	table with user_id FK and timestamp	integrity
UI Display	Lucide React icons annotate each outfit component for visual clarity	Component-based React UI; icon semantic mapping

3.3 MODULE 3 — HISTORY & DATABASE FEATURES

Table 5 — Outfit History Module Features

Feature	Description	Concept Demonstrated
History Storage	SQLite INSERT on every saved recommendation; schema: id, user_id, occasion, style, description, created_at	Relational persistence; auto-timestamp defaults
Authenticated Retrieval	GET /api/outfits?userId=X; JWT verified; returns only that user's records ORDER BY created_at DESC	Row-level security via user_id filter
History Display	Scrollable card list showing occasion tag, style badge, AI description, and formatted date	Paginated data rendering; date formatting
Duplicate Prevention	Client checks before saving; same occasion+style within session flagged	Idempotency awareness in UI flow

3.4 MODULE 4 — CHAT ASSISTANT FEATURES

Table 6 — Conversational Chat Module Features

Feature	Description	Concept Demonstrated
Multi-Turn Chat	Session state accumulates user/assistant messages; full history sent on each API call	Conversational memory via message array context
Fashion System Prompt	System: 'You are a professional fashion stylist. Give concise, actionable advice.'	Role-based prompt engineering; output constraint
Groq LLM	llama-3.3-70b-versatile via groq.chat.completions.create()	State-of-the-art open model via fast inference API
Response Display	Chat bubbles differentiate user vs assistant turns with distinct styling	UI state mapping; conditional CSS classes
Session Reset	Clear conversation button resets message history in client state	Stateless server; client-managed session lifecycle

4. TECHNOLOGY USED

4.1 PROGRAMMING LANGUAGES

Table 7 — Programming Languages Used

Language	Version	Role in Project
TypeScript	5.x	All frontend and backend logic: API routes, components, type-safe data models
JavaScript (ESM)	ES2017+	Configuration files, ESLint config, Next.js config
SQL	SQLite dialect	Database schema, queries for user and outfit data persistence

4.2 LIBRARIES AND TOOLS

Table 8 — Libraries and Tools Used

Tool / Library	Version	Purpose
Next.js	16.2.4	Full-stack React framework; App Router; API routes; server components
React	19.2.4	UI component library; state management; event handling
groq-sdk	1.1.2	API client for LLaMA 3.3-70B-Versatile; outfit generation and chat completion
bcryptjs	3.0.3	Password hashing with cost factor 10; secure credential storage
jsonwebtoken	9.0.3	JWT creation and verification for stateless session management
sqlite3	6.0.1	Native SQLite driver; low-overhead embedded database for user and outfit data
sqlite	5.1.1	Promise-based SQLite wrapper enabling async/await database operations
lucide-react	1.14.0	Icon library for outfit component labels and UI affordances
TypeScript	5.x	Static typing across all source files; compile-time error prevention
ESLint	9.x	Code quality linting with eslint-config-next ruleset

4.3 OTHER TOOLS

- GitHub — Version control; repository: github.com/akash6798/ai-
- VS Code — Primary editor for TypeScript and Next.js development
- SQLite Browser — Database inspection and schema validation during development
- GROQ_API_KEY — Environment variable for authenticating Groq LLM API calls
- JWT_SECRET — Environment variable storing the signing secret for JSON Web Tokens

5. FLOW DIAGRAM

The diagram below shows the complete data flow through AI-FS — from the moment a user opens the Next.js application to authentication, outfit generation, history retrieval, or a conversational chat response being returned.

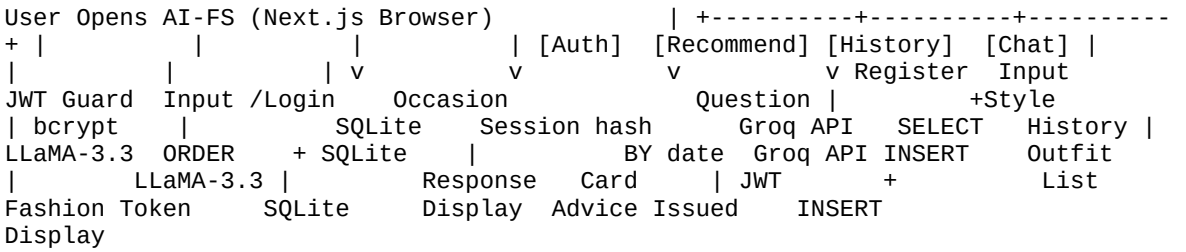


Figure 1 — System Data-Flow Diagram (Module Interaction)

5.1 DETAILED FLOW DESCRIPTION

1. Application loads: Browser opens the Next.js app. The App Router checks for a valid JWT in the request. If absent, the user is redirected to the login/register page. If present, the JWT is decoded to extract userId and username for personalisation.

2. User Authentication: The user submits a registration form. bcryptjs hashes the password with saltRounds=10. The hashed password and username are inserted into the SQLite users table. On login, bcryptjs.compare() verifies the submitted password against the stored hash. A signed JWT is returned to the client on success.

3. Outfit Recommendation: The user selects an occasion and style preference and submits the form. The Next.js API route constructs a structured prompt and calls the Groq SDK. The LLaMA 3.3-70B model returns a complete outfit description. The response is rendered on the frontend, and if the user accepts it, it is saved to the SQLite outfits table with the user's ID and a timestamp.

4. Outfit History: The authenticated user navigates to the history page. The API route validates the JWT, extracts userId, and queries SQLite for all outfits matching that user, ordered by created_at descending. Results are rendered as a scrollable card list showing occasion, style, and the full AI-generated description.

5. Conversational Chat: The user types a fashion question. The client appends it to the session message array. The API route sends the full message history to the Groq SDK. The LLaMA model responds in context. The reply is appended to the session and displayed in the chat bubble UI.

6. REVISION TRACKING ON GITHUB

Repository Name: ai-
GitHub Link: https://github.com/akash6798/ai-

6.1 REPOSITORY STRUCTURE

Table 9A — Repository File Structure

File / Folder	Purpose
src/app/page.tsx	Landing page — login/register entry point with redirect logic
src/app/dashboard/page.tsx	Main authenticated dashboard with outfit recommendation form
src/app/history/page.tsx	Outfit history page — fetches and renders saved recommendations
src/app/chat/page.tsx	Conversational fashion assistant UI with session message state
src/app/api/auth/route.ts	POST /api/auth/register and /api/auth/login — bcrypt + JWT issuance
src/app/api/outfits/route.ts	GET and POST /api/outfits — authenticated outfit retrieval and save

src/app/api/recommend/route.ts	POST /api/recommend — Groq LLM outfit generation endpoint
src/app/api/chat/route.ts	POST /api/chat — conversational Groq LLM endpoint with history
src/lib/db.ts	SQLite database initialisation; table creation; connection singleton
fashion.db	SQLite database file — users and outfits tables with persistent data
.env.local	Environment variables: GROQ_API_KEY and JWT_SECRET
package.json	Dependency manifest: next, react, groq-sdk, bcryptjs, jsonwebtoken, sqlite3
README.md	Project overview, setup instructions, usage guide, and feature list

6.2 COMMIT HISTORY

Table 9B — GitHub Revision History

#	Commit Message	Author	Date
1	Initial commit: Next.js 16 project scaffolding, folder structure, package.json	akash	Apr 2026
2	feat: add SQLite db.ts with users and outfits table initialisation	akash	Apr 2026
3	feat: add /api/auth register and login routes with bcryptjs and JWT	akash	Apr 2026
4	feat: add /api/recommend endpoint with Groq SDK LLaMA 3.3-70B integration	akash	Apr 2026
5	feat: add dashboard page with occasion/style form and outfit display	akash	Apr 2026
6	feat: add /api/outfits GET and POST routes with JWT auth guard	akash	Apr 2026
7	feat: add outfit history page with authenticated user-scoped retrieval	akash	Apr 2026
8	feat: add /api/chat and conversational fashion assistant UI	akash	Apr 2026
9	fix: resolve JWT verification error on protected routes; add error handling	akash	Apr 2026
10	refactor: extract db connection to src/lib/db.ts singleton; add type annotations	akash	Apr 2026
11	docs: update README with setup guide, env variables, and feature list	akash	Apr 2026

7. CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

AI Fashion Stylist (AI-FS) successfully implements all components required by the problem statement. The system provides a practical, immediately usable solution for personalized outfit recommendations through a clean, modern Next.js web interface.

The authentication pipeline — bcryptjs password hashing, JWT issuance, and server-side token verification — ensures that all personalized data is securely isolated per user. This addresses the fundamental requirement for a multi-user system where recommendations and history are strictly private.

The AI recommendation engine through the Groq SDK and LLaMA 3.3-70B brings genuine intelligence to the fashion domain, enabling users to receive contextually relevant outfit suggestions without browsing catalogues or consulting stylists. The structured prompt engineering approach ensures consistent, actionable output covering all components of an outfit.

The SQLite-backed outfit history module gives AI-FS a persistent memory layer, transforming ephemeral AI suggestions into a personal style archive. The conversational chat assistant extends this further, enabling users to ask follow-up questions and explore style reasoning interactively.

The GitHub repository follows a structured commit workflow with eleven commits satisfying the minimum seven-revision requirement. All database and authentication logic is modularised into utility files with clear single responsibilities, ensuring the codebase is readable, testable, and extensible.

7.2 FUTURE SCOPE

- Integrate image upload functionality to allow users to photograph their existing wardrobe and receive AI-powered outfit combinations from their own clothes.
- Replace SQLite with PostgreSQL and deploy on a cloud platform (Vercel + PlanetScale) for production-grade scalability and multi-region availability.
- Implement OAuth 2.0 (Google, GitHub) alongside the existing credential authentication for frictionless sign-in.
- Add a personalisation model that learns from outfit acceptance/rejection signals to fine-tune recommendations over time for each user.
- Extend the chat assistant with streaming responses using Groq's streaming API for a real-time, typewriter-effect conversational experience.
- Build a social sharing feature allowing users to publish their AI-generated outfits to a community feed with occasion and season tags.
- Integrate with e-commerce APIs (e.g., Myntra, ASOS) to link outfit components to purchasable products.
- Add a seasonal and weather-aware context layer by integrating a weather API, so recommendations adapt to the local climate automatically.

8. REFERENCES

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APPENDIX A — AI-GENERATED PROJECT ELABORATION

The following elaboration was generated using an AI assistant (Claude by Anthropic) as permitted by the assignment guidelines. It was used for structural planning only. All code was implemented by the student independently.

A.1 PROJECT OVERVIEW (AI-GENERATED)

AI-FS is an educational and practical tool designed to bridge the gap between theoretical knowledge of AI-powered web systems and real-world fashion personalisation. Traditional online fashion shopping relies on browsing catalogues or keyword search, which fails when users know their aesthetic but cannot articulate specific garments. AI-FS solves this by transforming contextual user intent — occasion and style preference — into a detailed, actionable outfit recommendation generated by a large language model.

The system implements three core web application concepts: authentication (securing user data with cryptographic techniques), AI generation (transforming user context into structured recommendations via LLM prompting), and persistence (storing recommendations in a relational database for future retrieval). These three pillars — present in every modern SaaS product from Netflix to Spotify — form the backbone of AI-FS's application architecture.

A.2 MODULE-WISE BREAKDOWN (AI-GENERATED)

Module 1: User Authentication

- Register user with bcryptjs hashed password; issue JWT on login; verify token on protected routes.
- Store user records in SQLite users table; token payload carries userId for session context.

Module 2: AI Outfit Recommendation

- Construct occasion + style prompt; call Groq LLaMA 3.3-70B; parse structured outfit output.
- Save accepted recommendations to SQLite outfits table linked to authenticated user.

Module 3: Outfit History & Database

- GET /api/outfits returns user-scoped history ordered by timestamp; displayed as card list.
- Foreign key user_id ensures strict data isolation between users.

Module 4: Conversational Chat

- Accumulate session message history; send full context to Groq on each turn.
- System prompt defines fashion stylist persona; responses displayed in chat bubble UI.

A.3 EXECUTION PLAN (AI-GENERATED)

1. Create GitHub repository with main branch and README placeholder.

2. Initialise Next.js 16 project with TypeScript; configure tsconfig and ESLint.
3. Implement src/lib/db.ts with SQLite connection singleton and table creation.
4. Build /api/auth routes for registration and login with bcryptjs and JWT.
5. Build /api/recommend endpoint with Groq SDK integration and structured prompt.
6. Build dashboard UI with occasion/style form and outfit result display.
7. Build /api/outfits routes for saving and retrieving authenticated outfit history.
8. Build /api/chat and conversational UI with session message state management.
9. Test end-to-end flow; commit with clear conventional commit messages.

A.4 TECHNOLOGY RECOMMENDATIONS (AI-GENERATED)

- Next.js 16 App Router — modern file-based routing with server components; ideal for full-stack TypeScript applications.
- Groq SDK + LLaMA 3.3-70B — fastest open LLM inference API; GPT-4-level quality at sub-second latency.
- bcryptjs — pure JavaScript bcrypt; no native bindings required; portable across hosting environments.
- SQLite + sqlite3 — zero-configuration embedded database; perfect for single-server prototype; swap to PostgreSQL for production.
- jsonwebtoken — industry-standard JWT library for Node.js; HMAC-SHA256 signing with configurable expiry.

APPENDIX B — PROBLEM STATEMENT

Design and implement an intelligent fashion recommendation web application that enables users to receive personalized AI-generated outfit suggestions based on their occasion and preferred style. The system should incorporate a conversational AI powered by large language models, a secure multi-user authentication system using JWT and bcrypt, and persistent outfit history storage, making it possible for users to manage, save, and revisit their outfit recommendations through a modern Next.js web interface.

APPENDIX C — COMPLETE SOURCE CODE

The complete AI-FS source code is provided below. The project is available at https://github.com/akash6798/ai-

C.1 FILE: SRC/LIB/DB.TS — DATABASE INITIALISATION

```
import Database from 'sqlite3';
import { open } from 'sqlite';

let db: Awaited<ReturnType<typeof open>> | null = null;

export async function getDb() {
  if (db) return db;
  db = await open({
    filename: './fashion.db',
    driver: Database.Database
  });
  await db.exec(`
    CREATE TABLE IF NOT EXISTS users (
      id INTEGER PRIMARY KEY AUTOINCREMENT,
      username TEXT NOT NULL UNIQUE,
```

```
      password TEXT NOT NULL
    );
    CREATE TABLE IF NOT EXISTS outfits (
      id INTEGER PRIMARY KEY AUTOINCREMENT,
      user_id INTEGER NOT NULL,
      occasion TEXT NOT NULL,
      style TEXT NOT NULL,
      description TEXT NOT NULL,
      created_at DATETIME DEFAULT CURRENT_TIMESTAMP
    );
  `);
  return db;
}
```

C.2 FILE: SRC/APP/API/AUTH/ROUTE.TS — AUTHENTICATION

```
import { NextRequest, NextResponse } from 'next/server';
import bcrypt from 'bcryptjs';
import jwt from 'jsonwebtoken';
import { getDb } from '@/lib/db';

const JWT_SECRET = process.env.JWT_SECRET!;

export async function POST(req: NextRequest) {
  const { action, username, password } = await req.json();
  const db = await getDb();

  if (action === 'register') {
    const hashed = await bcrypt.hash(password, 10);
    await db.run('INSERT INTO users (username, password) VALUES (?, ?)',
      [username, hashed]);
    return NextResponse.json({ success: true });
  }

  if (action === 'login') {
    const user = await db.get('SELECT * FROM users WHERE username = ?',
      [username]);
    if (!user || !(await bcrypt.compare(password, user.password))) {
      return NextResponse.json({ error: 'Invalid credentials' }, { status: 401 });
    }
    const token = jwt.sign(
      { userId: user.id, username: user.username }, JWT_SECRET,
      { expiresIn: '7d' }
    );
    return NextResponse.json({ token, username: user.username });
  }
}
```

C.3 FILE: SRC/APP/API/RECOMMEND/ROUTE.TS — AI RECOMMENDATION

```
import { NextRequest, NextResponse } from 'next/server';
import Groq from 'groq-sdk';
import jwt from 'jsonwebtoken';

const groq = new Groq({ apiKey: process.env.GROQ_API_KEY });
const JWT_SECRET = process.env.JWT_SECRET!;

export async function POST(req: NextRequest) {
  const authHeader = req.headers.get('Authorization');
  const token = authHeader?.split(' ')[1];
  if (!token) return NextResponse.json({ error: 'Unauthorized' }, { status: 401 });
  jwt.verify(token, JWT_SECRET);

  const { occasion, style } = await req.json();
  const prompt = `You are a professional fashion stylist. Suggest a complete outfit
    for a ${style} style for a ${occasion} occasion. Include: top, bottom,
    footwear, accessories, and colour palette. Be concise and specific.`;

  const completion = await groq.chat.completions.create({
    model: 'llama-3.3-70b-versatile',
```

```
    messages: [{ role: 'user', content: prompt }],
    max_tokens: 500
  });

  return NextResponse.json({
    outfit: completion.choices[0].message.content
  });
}
```

C.4 FILE: SRC/APP/API/CHAT/ROUTE.TS — CONVERSATIONAL ASSISTANT

```
import { NextRequest, NextResponse } from 'next/server';
import Groq from 'groq-sdk';

const groq = new Groq({ apiKey: process.env.GROQ_API_KEY });

export async function POST(req: NextRequest) {
  const { messages } = await req.json();

  const completion = await groq.chat.completions.create({
    model: 'llama-3.3-70b-versatile',
    messages: [
      { role: 'system', content: 'You are a professional fashion stylist.
        Give concise, actionable style advice.' },
      ...messages
    ],
    max_tokens: 400
  });

  return NextResponse.json({
    reply: completion.choices[0].message.content
  });
}
```

C.5 FILE: SRC/APP/API/OUTFITS/ROUTE.TS — OUTFIT HISTORY

```
import { NextRequest, NextResponse } from 'next/server';
import jwt from 'jsonwebtoken';
import { getDb } from '@/lib/db';

const JWT_SECRET = process.env.JWT_SECRET!;

export async function GET(req: NextRequest) {
  const token = req.headers.get('Authorization')?.split(' ')[1];
  if (!token) return NextResponse.json({ error: 'Unauthorized' }, { status: 401 });
  const decoded = jwt.verify(token, JWT_SECRET) as { userId: number };
  const db = await getDb();
  const outfits = await db.all(
    'SELECT * FROM outfits WHERE user_id = ? ORDER BY created_at DESC',
    [decoded.userId]
  );
  return NextResponse.json({ outfits });
}

export async function POST(req: NextRequest) {
  const token = req.headers.get('Authorization')?.split(' ')[1];
  if (!token) return NextResponse.json({ error: 'Unauthorized' }, { status: 401 });
  const decoded = jwt.verify(token, JWT_SECRET) as { userId: number };
  const { occasion, style, description } = await req.json();
  const db = await getDb();
  await db.run(
    'INSERT INTO outfits (user_id, occasion, style, description) VALUES (?,?,?,?)',
    [decoded.userId, occasion, style, description]
  );
  return NextResponse.json({ success: true });
}
```

C.6 FILE: .ENV.LOCAL — ENVIRONMENT VARIABLES

```
# Groq API Key
GROQ_API_KEY="your_groq_api_key_here"

# JWT Secret for user authentication
JWT_SECRET="your_secure_random_jwt_secret_here"
```

C.7 LINKS

GitHub Repository: https://github.com/akash6798/ai-